

Shawal Latif

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Education

Lahore University of Management Sciences (LUMS), Lahore, Pakistan August 2024–Present
M.S. in Artificial Intelligence
Selected coursework: AI on Edge Devices; Topics in Large Language Models; Human-Centered AI; Big Data Analytics.
COMSATS University Islamabad, Pakistan 2017–2021
B.S. in Electrical Engineering

Experience

Full-Stack AI Engineer, Awaaz-e-Sehat January 2025–Present
National AI Hub, LUMS | Supervisor: Dr. Maryam Mustafa

- Leading the **end-to-end system architecture** of a voice-first maternal-health assistant for low-literacy women, supported by the Gates Foundation-funded National AI Hub.
- Integrating speech processing, LLM inference, **retrieval-augmented generation (RAG)**, and backend services into the application’s conversational workflow, connecting spoken questions with maternal-health information.
- Evaluating language models for clinical correctness, linguistic fairness, and accessibility; contributing to clinician-informed benchmarking of **30+ models** to identify failure modes and guide model selection.
- Driving iterative field and usability testing with rural women in Pakistan, translating findings on literacy, shared-device use, and cultural context into concrete interface requirements and model-design decisions.
- Mentoring junior engineers and students in system design, LLM evaluation, and responsible AI, connecting technical implementation with users’ needs.

Research Intern Summer 2025
Tsinghua University | Supervisor: Prof. Chen Gao

- Developed LLM-agent architectures with memory, reflection, and planning for a multi-agent simulation framework studying information diffusion, relationship formation, and group coordination.
- Investigated behavioral consistency across extended agent interactions, using social simulation to examine the community-level implications of AI deployment.

Research Manuscripts

A Little Help, Closer to Home: Toward Accessible Conversational AI for Maternal Health in Pakistan
Submitted to ACM CHI 2027

- Compared **ten large and small language models** on **200 real maternal-health questions** from a WhatsApp voice bot, examining whether smaller models could support women in rural Pakistan where internet access is unreliable.
- Developed seven response-quality dimensions through interviews with four obstetricians and three UX experts; combined iterative qualitative coding with three obstetricians’ separate assessments of 36 responses.
- Identified Roman Urdu comprehension and linguistic precision as barriers to local deployment, informing planned fine-tuning, on-device deployment, and usability testing with women using the application.

Clinical Fairness Auditing of Frontier LLMs

Manuscript in preparation; target: ACM FAccT 2027

- Auditing Roman Urdu and code-mixed maternal-health responses for linguistic and intersectional disparities using Reduced Care Rate, Reduced Care Error Rate, and Triage Threshold Shift metrics.
- Designing a **sycophancy evaluation** to test whether models affirm or respectfully correct health misinformation attributed to trusted relatives or peers, and whether agreement-seeking reduces clinical accuracy.

Research Manuscripts (continued)

BenchMaterna: A Clinician-Validated Benchmark for Evaluating LLMs on Maternal Healthcare in the Global South

Manuscript in preparation; target: ACL 2027

- Benchmarking more than 30 proprietary, open-source, small, and biomedical language models using regional-language maternal-health queries and reference answers co-constructed with 30 clinical experts.
- Evaluating clinical correctness, cultural relevance, health-literacy accessibility, demographic fairness, harmful outputs, and helpfulness for underserved women in low-resource settings.

Projects

System-Aware Compression and Edge Deployment of LLaMA Models

Independent Research

- Built a compression pipeline to run **LLaMA-2-7B on memory-constrained CPUs**, combining pruning, LoRA recovery fine-tuning, SmoothQuant calibration, and GGUF quantization.
- Extended `llama.cpp` to read layer-specific attention and feed-forward dimensions from GGUF metadata, resolving runtime incompatibilities with non-uniformly pruned models without changing the model format.
- Designed sensitivity-based mixed-precision allocation under a fixed memory budget; benchmarked WikiText-2 perplexity, throughput, prefill and decoding latency, and peak memory to compare deployment trade-offs.
- Achieved **1.25× CPU throughput** with 30% pruning, increasing generation speed from 36.8 to 45.9 tokens/s. Separately, Q4_K_M quantization reduced storage from **25.10 to 3.41 GB** and peak memory from **14.99 to 3.75 GB**, with perplexity increasing from 7.18 to 8.52.

LLM Twin: End-to-End Production LLM System

Personal Project

- Built a **personalized assistant** that learns a user's writing style and answers from their content, with independently deployable feature, training, and inference pipelines orchestrated in ZenML.
- Implemented MongoDB ingestion, change data capture, and Bytewax streaming to keep Qdrant retrieval current; combined hybrid search, query expansion, and cross-encoder reranking to supply relevant context.
- Fine-tuned Llama 3.1 8B with LoRA/QLoRA and direct preference optimization; evaluated responses with LLM-as-a-judge and RAGAS, tracking experiments in Comet ML.
- Deployed containerized FastAPI services and a LangGraph RAG pipeline on **AWS SageMaker**; automated testing, deployment, and retraining with GitHub Actions, with Opik monitoring and data and concept drift detection.

Teaching and Mentoring

Graduate Teaching Assistant, Computer Science, LUMS

2024–Present

- Delivered tutorials and office hours for **200+ students across seven courses**: AI for Social Impact, Topics in Large Language Models, Vision-Language Models, Mathematics for AI, Probability and Statistics, Calculus II, and AI on Edge Devices.
- Led practical labs for AI on Edge Devices and Vision-Language Models; designed assignments and exam questions, graded student work, and provided feedback.
- Received the **Best TA Award for Mathematics for AI**.

Technical Expertise

Programming and Data Analysis: Python, SQL, Bash, JavaScript, MATLAB; pandas, NumPy.

Machine Learning and Adaptation: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, PEFT, TRL, Unsloth; supervised fine-tuning (SFT), LoRA/QLoRA, preference alignment (DPO/RLHF).

Retrieval and Agent Systems: LangChain, LangGraph, Sentence Transformers, Qdrant; vector and hybrid search, query expansion, self-querying, cross-encoder reranking, prompt engineering.

Compression and Inference: Pruning, SmoothQuant, sensitivity-based mixed precision, GPTQ, AWQ, GGUF; `llama.cpp`, vLLM, Text Generation Inference (TGI), TensorRT-LLM, ExLlamaV2; speculative decoding, Flash Attention, KV-cache optimization, continuous batching.

Evaluation and MLOps: LLM-as-a-judge, RAGAS, Comet ML, Opik, ZenML; Docker, GitHub Actions (CI/CD/CT), automated retraining, prompt monitoring, data and concept drift detection.

Cloud, Backend, and Data Engineering: AWS (SageMaker, ECR, S3, CloudFormation, IAM); FastAPI, Pydantic, Poetry, REST APIs, microservices; MongoDB, Bytewax, Selenium, ETL pipelines, streaming, change data capture (CDC).